Envelope–Santaló extension for extremal general affine surface area

Candidate full result for arXiv:2103.00294

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Status. Candidate full result, likely valid, pending specialist review. The intended concave-side program is completed for a new nonmonotone class. Two printed claims in the source require correction, so the theorem below proves the outer statements directly rather than using the source's polarity argument.

1 Problem and outcome

Hoehner studies extremal general affine surface areas under global monotonicity of $\varphi(t)/\sqrt{t}$ and asks for the results of the article for a richer class than $\mathrm{Conc}^-(0,\infty)\cup\mathrm{Conc}^+(0,\infty)$ [1]. The exact source prompt is reproduced below.

> "Ultimately, it is an open problem to obtain the results of this article for a richer class of functions than $\mathrm{Conc}^-(0,\infty)\cup\mathrm{Conc}^+(0,\infty)$."

function $G_n(t)=\psi(t^{n+1})$ for $t\in(0,\infty)$ is convex. Then

$$\mathrm{as}_\psi(K)\geq \mathrm{as}_\psi(\mathrm{vrad}(K)B_n).$$

Moreover, if $G_n(\cdot)$ is strictly convex, then equality holds if and only if K is an origin-symmetric ellipsoid.

Thus, one could remove the assumption that the bodies in $\mathscr{C}_{\mathrm{in}}(K)$ or $\mathscr{C}_{\mathrm{out}}(K)$ have centroid at the origin by using Lemmas 7.1 and 7.2 in all of the proofs instead of Lemmas 2.1 and 2.2, but the downside is then one captures fewer functions from $\mathrm{Conc}(0,\infty)$ and $\mathrm{Conv}(0,\infty)$. Furthermore, the assumption in the definitions of $\mathscr{C}_{\mathrm{in}}(K)$ and $\mathscr{C}_{\mathrm{out}}(K)$ that the bodies have centroid at the origin is tacitly assumed in the definition of the extremal L_p affine surface areas in [18]. This is because the definition (39) of L_p affine surface area is stated for convex bodies with centroid at the origin. Ultimately, it is an open problem to obtain the results of this article for a richer class of functions than $\mathrm{Conc}^-(0,\infty)\cup\mathrm{Conc}^+(0,\infty)$.

We introduce an *interior-peaked* class, prove existence, attainment, Hausdorff continuity, sharp affine-isoperimetric envelopes, and both inner and outer Blaschke–Santaló inequalities, and determine the exact equality conditions. The class contains

$$\varphi(t)=\log(1+t),$$

whose quotient by $\sqrt{t}$ is nonmonotone, so it is genuinely outside both source classes. Equality is necessarily different from the source theory: this example has many non-ellipsoidal equality bodies.

The proof also exposes two independent issues in the published statements. First, the claimed "equality iff ellipsoid" assertion in the source Blaschke–Santaló theorem fails on a scaled ball for the paper's own arctangent example. Second, centroid zero is not preserved by polarity, so the asserted bijection between centered inner and centered outer competitors is false. Neither issue affects the direct proofs below.

2 Definitions and source inputs

Let B_n be the Euclidean unit ball in $\mathbb{R}^n$ and let

$$\mathrm{vrad}(K) = \left(\frac{|K|}{|B_n|}\right)^{1/n}.$$

All ambient bodies contain the origin in their interiors. Write

$$\mathcal{C}_{\mathrm{in}}(K) = \{C \subset K : C \text{ is a convex body and } g(C) = 0\},$$

$$\mathcal{C}_{\mathrm{out}}(K) = \{D \supset K : D \text{ is a convex body and } g(D) = 0\}.$$

For $\varphi \in \mathrm{Conc}(0, \infty)$, put

$$\mathrm{IS}_\varphi(K) = \sup_{C \in \mathcal{C}_{\mathrm{in}}(K)} \mathrm{as}_\varphi(C).$$

Define $\varphi^*(t) = t\varphi(1/t)$ and retain the source's original outer functional

$$\mathrm{OS}_{\varphi^*}(K) = \sup_{D \in \mathcal{C}_{\mathrm{out}}(K)} \mathrm{as}_{\varphi^*}(D).$$

The involution preserves $\mathrm{Conc}(0, \infty)$.

We use three standard facts from Ludwig's general affine surface-area theory [2]. For a centered convex body C and $\theta \in \mathrm{Conc}(0, \infty)$,

$$\mathrm{as}_\theta(C) \le \mathrm{as}_\theta(\mathrm{vrad}(C)B_n), \tag{1}$$

with equality for strictly increasing θ only when C is an ellipsoid; as_θ is upper semicontinuous under Hausdorff convergence; and

$$\mathrm{as}_\theta(rB_n) = n|B_n|\frac{\theta(r^{-2n})}{\sqrt{r^{-2n}}}. \tag{2}$$

Set

$$h_\varphi(t) = \frac{\varphi(t)}{\sqrt{t}}, \qquad L_\varphi = \limsup_{t\to\infty} h_\varphi(t), \qquad H_\varphi(a) = \sup_{t \ge a} h_\varphi(t).$$

We call φ *tail-gapped* if

$$L_\varphi < \infty \quad \text{and} \quad H_\varphi(a) > L_\varphi \quad \text{for every } a > 0. \tag{3}$$

It is *interior-peaked* if it is tail-gapped and

$$M_\varphi := \sup_{t>0} h_\varphi(t) = H_\varphi(1). \tag{4}$$

Thus a global radial maximum is accessible at a finite parameter $t \ge 1$. Indeed, $H_\varphi(1) > L_\varphi$ implies that the supremum is attained. Let

$$\mathcal{A}_\varphi = \{t > 0 : h_\varphi(t) = M_\varphi\}. \tag{5}$$

The basic identity for the dual quotient is

$$\frac{\varphi^*(t)}{\sqrt{t}} = \sqrt{t}\,\varphi(1/t) = h_\varphi(1/t). \tag{6}$$

3 Idea of the proof

Affine isoperimetry reduces both extremal problems to the same one-dimensional tail envelope. An inner competitor of volume radius r uses the parameter r^{-2n}, whereas an outer φ^* competitor uses r^{2n} by (6). Finite tail limsup bounds both suprema. The strict tail gap is the compactness mechanism: it places a finite-radius ball strictly above the value available to an inner sequence collapsing to zero volume or an outer sequence escaping to infinite volume. Blaschke selection then gives both maximizers.

For continuity, concavity supplies a two-sided estimate under dilation; this is stronger than the one-sided estimate used in the source. Finally, the interior-peak condition says that every radial envelope is bounded by the value already accessible at the unit ball. It therefore proves both Blaschke–Santaló products factor by factor, without the source's invalid polar-centroid identification. Equality can then be read off from equality in Ludwig's affine isoperimetric inequality and membership in the maximizing set $\mathcal{A}_\varphi$.

4 Two-sided dilation control

The next elementary estimate is useful because it gives continuity for both supremal functionals without polarizing their competitors.

Proposition 1. *If $\theta \in \mathrm{Conc}(0,\infty)$, D is a convex body containing the origin, and $c \geq 1$, then*

$$c^{-n}\,\mathrm{as}_\theta(D) \leq \mathrm{as}_\theta(cD) \leq c^n\,\mathrm{as}_\theta(D). \tag{7}$$

Consequently, whenever the four quantities are finite,

$$c^{-n}\,\mathrm{IS}_\theta(K) \leq \mathrm{IS}_\theta(cK) \leq c^n\,\mathrm{IS}_\theta(K), \tag{8}$$

$$c^{-n}\,\mathrm{OS}_\theta(K) \leq \mathrm{OS}_\theta(cK) \leq c^n\,\mathrm{OS}_\theta(K). \tag{9}$$

Proof. Every positive concave function on $(0,\infty)$ with value 0 at the origin is nondecreasing; otherwise a negative secant slope and concavity would force the function eventually to become negative. The change of scale in the defining integral gives

$$\mathrm{as}_\theta(cD) = c^n \int_{\partial D} \theta(c^{-2n} q_D(x))\langle x, N_D(x)\rangle\, d\mu_D(x),$$

where q_D is the centro-affine curvature ratio. For $\lambda = c^{-2n} \in (0,1]$, monotonicity and concavity at the origin give

$$\lambda\theta(u) \leq \theta(\lambda u) \leq \theta(u).$$

This proves (7). The maps $C \mapsto cC$ and $D \mapsto cD$ are bijections between the corresponding inner and outer competitor families, so taking suprema proves (8) and (9). □

5 Full radial-envelope theorem

Theorem 2. *Let $\varphi \in \mathrm{Conc}(0,\infty)$ be tail-gapped. For every convex body K containing the origin in its interior:*

1. *$\mathrm{IS}_\varphi(K)$ and $\mathrm{OS}_{\varphi^*}(K)$ are positive and finite, are attained by centered convex bodies, and depend continuously on K in the Hausdorff metric;*

2. the sharp envelope inequalities hold:

$$\mathrm{IS}_\varphi(K) \le n|B_n| H_\varphi(\mathrm{vrad}(K)^{-2n}), \tag{10}$$

$$\mathrm{OS}_{\varphi^*}(K) \le n|B_n| H_\varphi(\mathrm{vrad}(K)^{2n}); \tag{11}$$

3. equality holds in (10) *and* (11) *for every centered Euclidean ball* $K = RB_n$.

Proof. The inner assertion is included for completeness. If $C \in \mathcal{C}_{\mathrm{in}}(K)$ and $s_C = \mathrm{vrad}(C)^{-2n}$, then $s_C \ge \mathrm{vrad}(K)^{-2n}$. Equations (1) and (2) give

$$\mathrm{as}_\varphi(C) \le n|B_n| h_\varphi(s_C) \le n|B_n| H_\varphi(\mathrm{vrad}(K)^{-2n}).$$

This proves finiteness and (10). If a maximizing sequence collapsed to a lower-dimensional set, then $s_C \to \infty$ and its limiting value would be at most $n|B_n|L_\varphi$. A centered ball $\rho B_n \subset K$ and the strict tail gap supply a smaller centered ball in K with value strictly larger than $n|B_n|L_\varphi$. Collapse is impossible. Blaschke selection, centroid continuity, and upper semicontinuity then give an inner maximizer.

For the outer assertion, let $D \in \mathcal{C}_{\mathrm{out}}(K)$ and put $u_D = \mathrm{vrad}(D)^{2n}$. By (6),

$$\mathrm{as}_{\varphi^*}(D) \le n|B_n| h_\varphi(u_D) \le n|B_n| H_\varphi(\mathrm{vrad}(K)^{2n}),$$

which proves finiteness and (11).

Let $S = \mathrm{OS}_{\varphi^*}(K)$. Choose R with $K \subset RB_n$. The strict gap $H_\varphi(R^{2n}) > L_\varphi$ supplies $u \ge R^{2n}$ with $h_\varphi(u) > L_\varphi$. Hence the centered ball $u^{1/(2n)}B_n \supset K$ shows

$$S > n|B_n|L_\varphi. \tag{12}$$

Let D_j be an outer maximizing sequence. Its volumes are bounded: an unbounded-volume subsequence would have $u_{D_j} \to \infty$ and therefore, by the preceding affine-isoperimetric estimate, limsup at most $n|B_n|L_\varphi$, contradicting (12). Fix $\rho B_n \subset K$. Since every D_j contains ρB_n, the cone with apex $x \in D_j$ and base $\rho B_n \cap x^\perp$ lies in D_j, so

$$|D_j| \ge \frac{|x|\rho^{n-1}|B_{n-1}|}{n}.$$

The volume bound therefore gives a uniform diameter bound. Blaschke selection produces a Hausdorff limit $D_0 \supset K$; it is full-dimensional, has centroid zero, and upper semicontinuity gives $S \le \mathrm{as}_{\varphi^*}(D_0) \le S$. Thus the outer supremum is attained.

For $K = RB_n$, every parameter $s \ge R^{-2n}$ is realized by the inner ball $s^{-1/(2n)}B_n$, and every $u \ge R^{2n}$ is realized by the outer ball $u^{1/(2n)}B_n$. This proves both sharpness statements.

It remains to prove continuity. Both functionals are monotone under inclusion, with IS increasing and OS decreasing. If $K_j \to K$, then there are $c_j, d_j \downarrow 1$ such that $K_j \subset c_jK$ and $K \subset d_jK_j$. Inclusion monotonicity and Proposition 1 give

$$d_j^{-n}\,\mathrm{IS}_\varphi(K) \le \mathrm{IS}_\varphi(K_j) \le c_j^n\,\mathrm{IS}_\varphi(K)$$

and

$$c_j^{-n}\,\mathrm{OS}_{\varphi^*}(K) \le \mathrm{OS}_{\varphi^*}(K_j) \le d_j^n\,\mathrm{OS}_{\varphi^*}(K).$$

Both squeeze limits prove Hausdorff continuity. □

6 Blaschke–Santaló theorem and exact equality

Theorem 3. *Let φ be interior-peaked. Then for every convex body K containing the origin in its interior,*

$$\mathrm{IS}_{\varphi}(K) \le \mathrm{IS}_{\varphi}(B_n) = n|B_n|M_{\varphi}, \tag{13}$$

$$\mathrm{OS}_{\varphi^*}(K) \le \mathrm{OS}_{\varphi^*}(B_n) = n|B_n|M_{\varphi}. \tag{14}$$

Consequently,

$$\mathrm{IS}_{\varphi}(K)\,\mathrm{IS}_{\varphi}(K^{\circ}) \le \mathrm{IS}_{\varphi}(B_n)^2, \tag{15}$$

$$\mathrm{OS}_{\varphi^*}(K)\,\mathrm{OS}_{\varphi^*}(K^{\circ}) \le \mathrm{OS}_{\varphi^*}(B_n)^2. \tag{16}$$

No centroid assumption on K is needed.

If φ is strictly increasing, equality in (13) holds if and only if K contains a centered ellipsoid E such that

$$\mathrm{vrad}(E)^{-2n} \in \mathcal{A}_{\varphi}. \tag{17}$$

If φ^* is strictly increasing, equality in (14) holds if and only if K is contained in a centered ellipsoid D such that

$$\mathrm{vrad}(D)^{2n} \in \mathcal{A}_{\varphi}. \tag{18}$$

Thus equality in either product inequality holds exactly when the corresponding containment condition holds for both K and K°.

Proof. By definition of the interior-peaked class, $H_{\varphi}(a) \le M_{\varphi} = H_{\varphi}(1)$ for every $a > 0$. Theorem 2, applied to K and to B_n, proves (13) and (14). Applying each universal bound to K and K° proves the two product inequalities.

Suppose φ is strictly increasing and equality holds in (13). Let $C \in \mathcal{C}_{\mathrm{in}}(K)$ be an inner maximizer from Theorem 2. The chain

$$n|B_n|M_{\varphi} = \mathrm{as}_{\varphi}(C) \le n|B_n|h_{\varphi}(\mathrm{vrad}(C)^{-2n}) \le n|B_n|M_{\varphi}$$

is equality throughout. Ludwig's equality condition makes C an ellipsoid, and its parameter belongs to $\mathcal{A}_{\varphi}$. The converse follows by using such an ellipsoid as a competitor. The outer proof is identical with an outer maximizer, (6), and strict increase of φ^*. Finally, two positive numbers each bounded by the same positive constant have maximal product exactly when both attain that constant. This gives the product equality criteria. □

7 The logarithmic class member and non-rigid equality

Corollary 4. *Let $\varphi(t) = \log(1+t)$. There is a unique $t_* > 1$ satisfying*

$$\log(1+t_*) = \frac{2t_*}{1+t_*}, \tag{19}$$

and $t_* = 3.921553634568\ldots$. The function φ is interior-peaked, belongs to neither source monotonicity class, and $\mathcal{A}_{\varphi} = \{t_*\}$. Both φ and φ^* are strictly increasing, so all conclusions and equality conditions of Theorem 3 apply.

Put $r_* = t_*^{-1/(2n)} < 1$*. Then*

$$\mathrm{IS}_\varphi(K) = \mathrm{IS}_\varphi(B_n) \iff K \textit{ contains a centered ellipsoid of volume radius } r_*,$$
$$\mathrm{OS}_{\varphi^*}(K) = \mathrm{OS}_{\varphi^*}(B_n) \iff K \textit{ is contained in a centered ellipsoid of volume radius } r_*^{-1}.$$

For $n \geq 2$ *and*

$$0 < \varepsilon < r_*^{-1} - 1, \tag{20}$$

the capsule

$$K_\varepsilon = B_n + \varepsilon[-e_1, e_1]$$

is not an ellipsoid, yet equality holds for K_ε *in both product inequalities* (15) *and* (16)*.*

Proof. Differentiation shows that the sign of $h'_\varphi(t)$ is the sign of

$$q(t) = \frac{2t}{1+t} - \log(1+t).$$

Since $q'(t) = (1-t)/(1+t)^2$, the function q increases on $(0,1)$, decreases on $(1,\infty)$, and tends to $-\infty$. It has the unique positive zero $t_* > 1$, so h_φ increases up to t_* and decreases afterward. Moreover, $h_\varphi(t) \to 0$ at both endpoints. Hence φ is interior-peaked, globally nonmonotone, and has the stated unique maximizer. The function φ is strictly increasing, while

$$(\varphi^*)'(t) = \log(1 + 1/t) - \frac{1}{1+t} > 0$$

because $\log(1+x) > x/(1+x)$ for $x > 0$.

The equality characterizations now follow from Theorem 3. For the explicit capsule,

$$r_* B_n \subset B_n \subset K_\varepsilon \subset (1+\varepsilon)B_n \subset r_*^{-1} B_n.$$

Polarity reverses these inclusions, so the same two-sided containment holds for K_ε°. Thus both K_ε and its polar satisfy both equality criteria. The capsule has nontrivial cylindrical boundary segments and is not an ellipsoid. □

8 Two corrections to the source paper

8.1 The printed ellipsoid equality claim is false

The source's Theorem 6.4 asserts, under strict increase of φ, that equality in its Blaschke–Santaló inequality holds if and only if K is an ellipsoid. Its own family

$$\varphi_m(t) = \arctan(t^{1/m}), \qquad m \geq 2,$$

contradicts the "if" direction. The source proves that h_{φ_m} is strictly decreasing and that

$$\varphi_m(s)\varphi_m(1/s) < \varphi_m(1)^2 \qquad (s \neq 1). \tag{21}$$

Indeed, if $a = \arctan(s^{1/m})$, then the left side is $a(\pi/2 - a)$, whose unique maximum occurs at $a = \pi/4$, equivalently $s = 1$.

Choose the ellipsoid $K = rB_n$ with $r \neq 1$ and set $s = r^{-2n}$. Since h_{φ_m} is decreasing, the sharp ball formula gives

$$\mathrm{IS}_{\varphi_m}(rB_n)\,\mathrm{IS}_{\varphi_m}(r^{-1}B_n) = (n|B_n|)^2\varphi_m(s)\varphi_m(1/s) < (n|B_n|)^2\varphi_m(1)^2 = \mathrm{IS}_{\varphi_m}(B_n)^2.$$

Thus a scaled Euclidean ball is an ellipsoid but is not an equality case. The exact envelope equality criterion in Theorem 3 avoids this homogeneity error.

8.2 Centered competitor families are not polar duals

The source states that $C \in \mathcal{C}_{\text{in}}(K)$ if and only if $C^\circ \in \mathcal{C}_{\text{out}}(K^\circ)$. Under the printed definitions this would require $g(C) = 0$ if and only if $g(C^\circ) = 0$, which is false.

An exact planar counterexample is the quadrilateral

$$C = \operatorname{conv}\left\{ \frac{(-7,-4)}{9}, \frac{(11,-4)}{9}, \frac{(2,5)}{9}, \frac{(-7,5)}{9} \right\}.$$

The polygon-centroid formula gives $g(C) = (0,0)$. Directly intersecting the four polar half-planes gives

$$C^\circ = \operatorname{conv}\left\{(0,-9/4), (9/7, 9/7), (0, 9/5), (-9/7, 0)\right\},$$

whose centroid is

$$g(C^\circ) = (0, 9/140) \neq 0.$$

Taking $K = C$ disproves the asserted competitor-family equivalence and hence the extremal polarity identity derived from it. This is why Theorems 2 and 3 treat the original outer functional directly.

9 Scope, verification, and novelty

Theorem 3 supplies the complete intended chain—existence, attainment, continuity, sharp affine bounds, inner and outer Blaschke–Santaló inequalities, and exact equality—for a concrete third class outside the source monotonicity union. It does not assert that every tail-gapped function satisfies the normalized product inequality; the global peak condition is precisely what makes both original functionals uniformly bounded by their unit-ball values.

The proof was checked against collapse and escape of maximizing sequences, centroid preservation under Hausdorff limits, both directions of the dilation estimate, and equality in every inequality of the maximizing chain. The script `code/check_examples.py` verifies the reported logarithmic root, strictness of the arctangent scaled-ball example, and both rational polygon centroids. The computations are not used in place of analytic proofs.

A bounded novelty search covered the four local run indexes, exact title and phrase searches, the arXiv record, the journal version, and works listed as citing the paper through 19 July 2026. No erratum, later solution of the open problem, radial-envelope theorem, or published notice of the two corrections was found. Absolute novelty is not certified.

Human review recommendation. Send to a convex-geometric analyst. The highest-value checks are the direct outer compactness proof, the two-sided dilation identity for nonsmooth bodies, and the interpretation of "full" as the complete intended concave-side program for a genuinely new class rather than a verbatim preservation of the source's two incorrect statements.

References

[1] S. Hoehner, *Extremal general affine surface areas*, J. Math. Anal. Appl. 505 (2022), 125506; arXiv:2103.00294.

[2] M. Ludwig, *General affine surface areas*, Adv. Math. 224 (2010), 2346–2360.